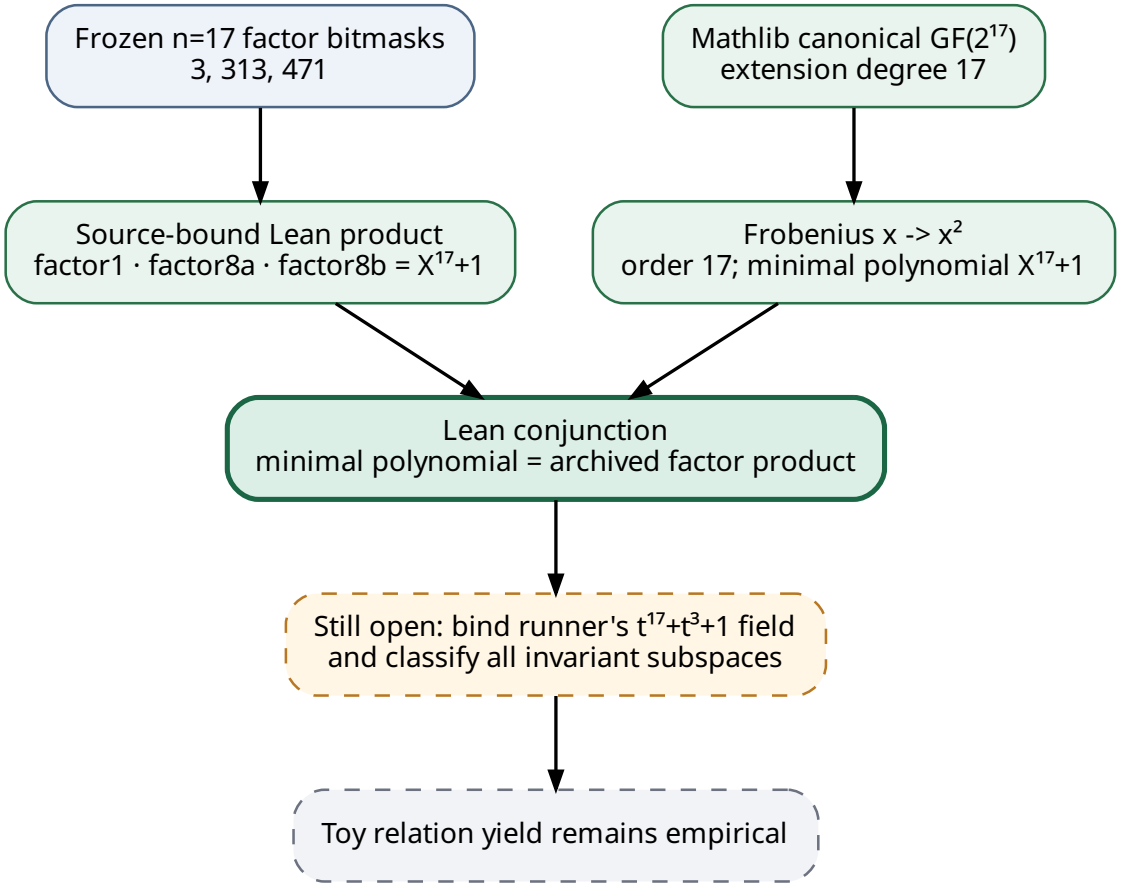
Canonical GF(2^{17}) Frobenius and BINSTD factors

Validation audit, 22 October 2026

Frozen source. The BINSTD n=17 dimension files record bitmask factors 3, 313 and 471 of $X^{17}-1$ over $\mathbf{F}_2$. Earlier Lean theorems bind those exact coefficients, prove their product is $X^{17}+1$ and prove both degree-eight factors irreducible.

New theorem. For the canonical Mathlib field GF(2^{17}), the $\mathbf{F}_2$-linear Frobenius $x \mapsto x^2$ has order 17 and minimal polynomial equal to the archived three-factor product. The proof instantiates Mathlib's finite-field Frobenius theorems and rewrites $X^{17}-1 = X^{17}+1$ in characteristic two. The formal task targets both identities. Its verification receipt reports a passing full build, standard axioms only across 630 audited theorems, and a passing target audit under Lean 4.35.0-rc1. Independent semantic review remains pending.

Boundary. The archived polynomial-basis runner uses $t^{17}+t^3+1$. Its concrete field, Frobenius implementation and subspace constructor are not identified with this canonical model in Lean. Exhaustive invariant-subspace classification remains open. No empirical yield, performance or security claim follows.


Frozen n=17 factor bitmasks
3, 313, 471
Mathlib canonical GF(2^17)
extension degree 17
Source-bound Lean product
factor1 · factor8a · factor8b = X^17+1
Frobenius x -> x^2
order 17; minimal polynomial X^17+1
Lean conjunction
minimal polynomial = archived factor product
Still open: bind runner's t^17+t^3+1 field
and classify all invariant subspaces
Toy relation yield remains empirical